Unitree Week 37

Presentation by Mario Ortiz

This Week

- Upper Body Control and Onboard PC Access for Unitree EDU

This Week

- Unitree currently provides R1-specific SDK examples for controlling the arms and accessing built-in arm actions.
- They have an **R1 Arm Action** Example, as well as an **R1 Arm SDK** Example.
  - R1 Arm Action:” queries and invokes built-in upper-body actions
  - R1 Arm SDK Example: Directly commands the robot’s arm joints.

r1_arm_action_example.cpp

```
12      * Usage:
13      *   r1_arm_action_example <networkInterface> --list
14      *   r1_arm_action_example <networkInterface> --id <action_id>
15      *   r1_arm_action_example <networkInterface> --name <action_name>
16      *   r1_arm_action_example <networkInterface> --stop
```

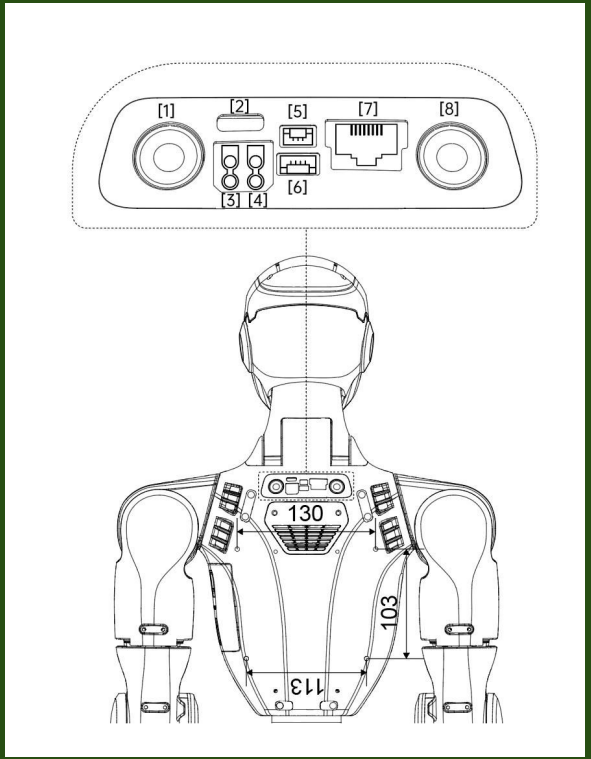
- Types of commands to use the script.

Connecting to the Unitree's Computer

- In order to use these SDK commands, we need to connect to the Unitree R1's software interface, via the computer.
  - This is done by connecting the machine to robot's internal network over Ethernet and then using SSH.
  - IP addresses used by the R1 EDU (from documentation):
    - Robot motion controller (PC1): 192.168.123.161 (Internal motion control)
    - EDU Development Computer (PC2): 192.168.123.164 (Software Development)

Step A: Connect a computer to the robot

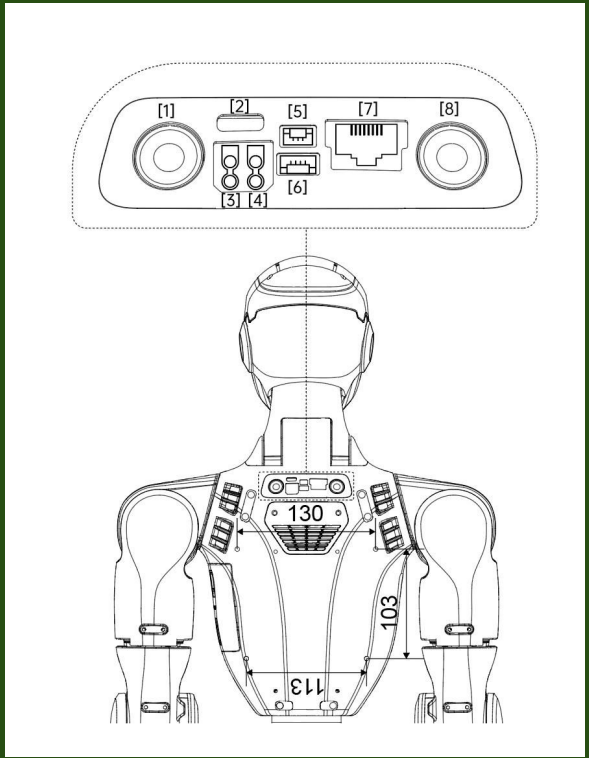
1. Connect an Ethernet cable from your computer to the R1's designated development network port.
2. Power on the R1 following its documented startup procedure, keeping it securely on the bracket.



Step B: Configure your Windows Computer

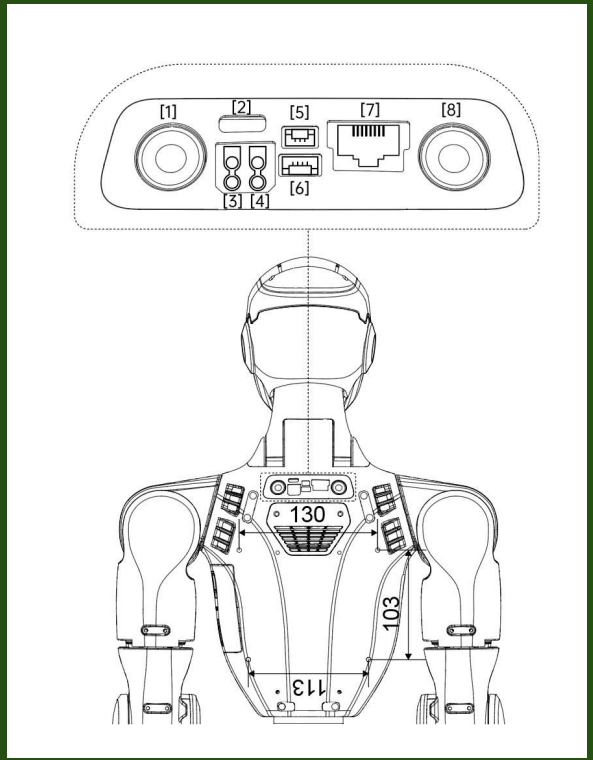
You can use Windows to ():

- Settings → Network & Ethernet → IP assignment → Edit
- In “Edit”, choose Manual → Enable IPv4 → then enter settings for:
  - IP address, Subnet mask, Gateway, DNS
- (must ensure the robot’s network (last 2 digits in IP address)



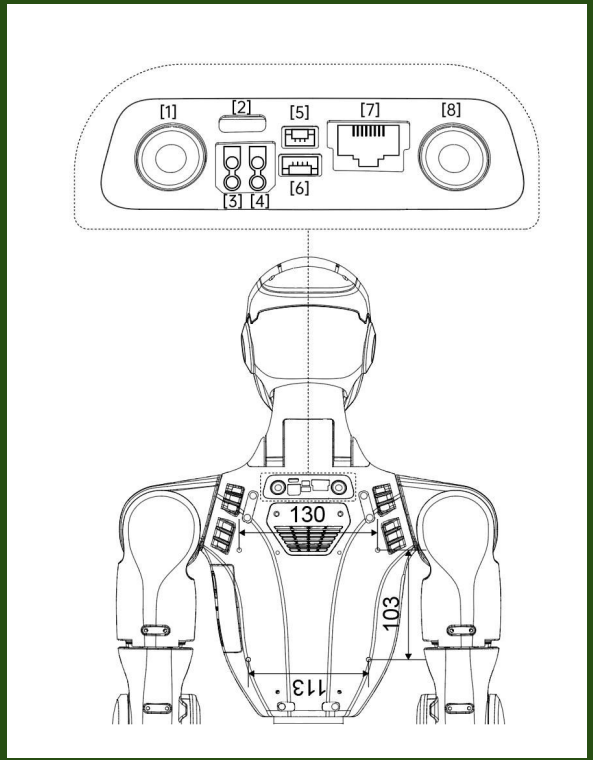
Step C: Test the connection

- On Windows, these commands will help test the connection:
- ping “Software Development IP; ex. 192.168.123.164”
- Check whether SSH is accessible:
  - Test-NetConnection 192.168.123.164 -Port 22



Step C: Test the connection

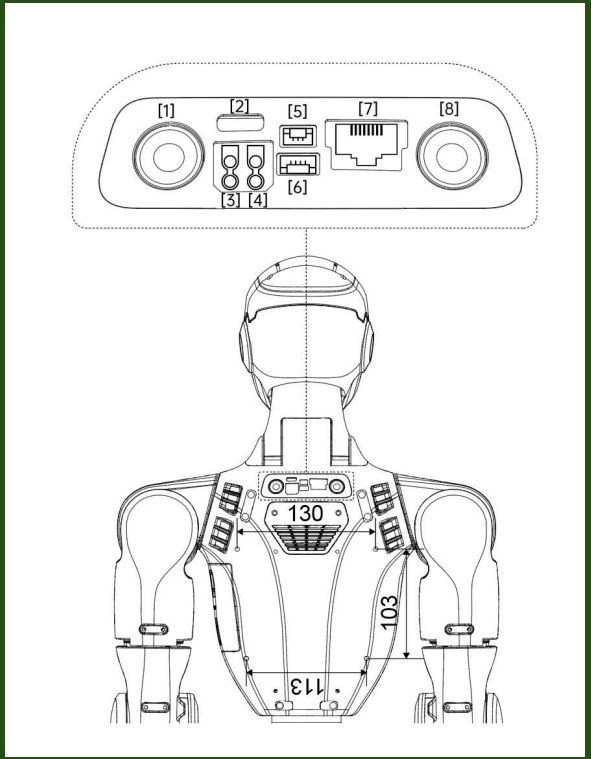
- Attempt to connect:
  - ssh unitree@a92.168.123.164
  - Note the publicly documented factory credentials are
    - Username: unitree
      Password: 123
- Operating system, architecture, and python version on the R1:
  hostnamect1
  cat /etc/os-release
  uname -a
  python3 --version



Step C: Test the connection

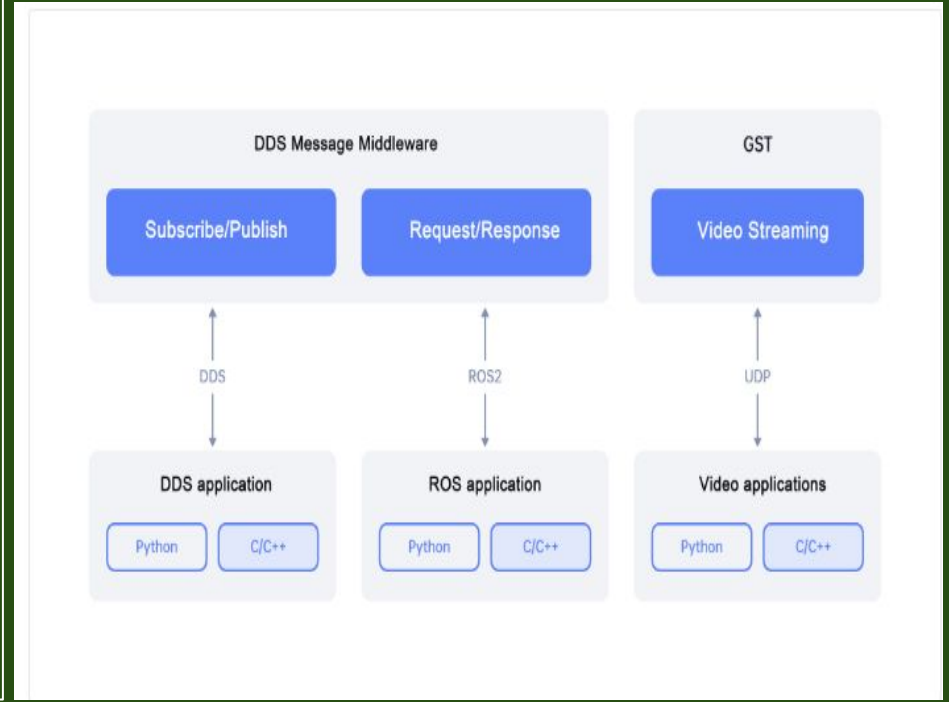
- Commands to show installed development environments and running services:

```
ls -la /home
ls -la /opt
systemct1 --type=service
--state=running --no-pager
command -v ros2
command -v docker
command -v cmake
```



DDS

- Note that the Unitree uses DDS as the intermediary software between the operating system and the higher-level application software.
- It uses “subscribe/publish” and “Request/Response” for sending messages



Connecting Ethernet Cable from R1 to Laptop

```
PS C:\Users\Mario> Get-NetAdapter

Name                      InterfaceDescription                    ifIndex Status       MacAddress
----                      --------------------                    ------- ------       ----------
Bluetooth Network Conn... Bluetooth Device (Personal Area Netw...      18 Disconnected 34-6F-24-52...
Ethernet                  Realtek PCIe GbE Family Controller           16 Up           50-EB-F6-47...
Wi-Fi                     MediaTek Wi-Fi 6 MT7921 Wireless LAN...       7 Up           34-6F-24-52...


PS C:\Users\Mario>
```

```
PS C:\Users\Mario> Get-NetIPInterface -InterfaceAlias "Ethernet" -AddressFamily IPv4

ifIndex InterfaceAlias                  AddressFamily NlMtu(Bytes) InterfaceMetric Dhcp     Connectio
                                                                                   nState
------- --------------                  ------------- ------------ --------------- ----     ---------
16      Ethernet                        IPv4                  1500              25 Enabled  Connected


PS C:\Users\Mario> New-NetIPAddress -InterfaceAlias "Ethernet" -IPAddress "192.168.123.99" -PrefixLeng
th 24


IPAddress         : 192.168.123.99
InterfaceIndex    : 16
InterfaceAlias    : Ethernet
AddressFamily     : IPv4
Type              : Unicast
PrefixLength      : 24
PrefixOrigin      : Manual
SuffixOrigin      : Manual
AddressState      : Tentative
ValidLifetime     : Infinite ([TimeSpan]::MaxValue)
PreferredLifetime : Infinite ([TimeSpan]::MaxValue)
SkipAsSource      : False
PolicyStore       : ActiveStore

IPAddress         : 192.168.123.99
InterfaceIndex    : 16
InterfaceAlias    : Ethernet
AddressFamily     : IPv4
Type              : Unicast
PrefixLength      : 24
PrefixOrigin      : Manual
SuffixOrigin      : Manual
AddressState      : Invalid
ValidLifetime     : Infinite ([TimeSpan]::MaxValue)
PreferredLifetime : Infinite ([TimeSpan]::MaxValue)
SkipAsSource      : False
PolicyStore       : PersistentStore
```

```
PS C:\Users\Mario> Get-NetIPAddress -InterfaceAlias "Ethernet" -AddressFamily IPv4


IPAddress         : 192.168.123.99
InterfaceIndex    : 16
InterfaceAlias    : Ethernet
AddressFamily     : IPv4
Type              : Unicast
PrefixLength      : 24
PrefixOrigin      : Manual
SuffixOrigin      : Manual
AddressState      : Preferred
ValidLifetime     : Infinite ([TimeSpan]::MaxValue)
PreferredLifetime : Infinite ([TimeSpan]::MaxValue)
SkipAsSource      : False
PolicyStore       : ActiveStore
```

Assigning the static IP address to that of the R1.

Testing Connection to the R1

```
Ping statistics for 192.168.123.164:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
PS C:\Users\Mario> ping 192.168.123.164

Pinging 192.168.123.164 with 32 bytes of data:
Reply from 192.168.123.164: bytes=32 time<1ms TTL=64
Reply from 192.168.123.164: bytes=32 time<1ms TTL=64
Reply from 192.168.123.164: bytes=32 time<1ms TTL=64
Reply from 192.168.123.164: bytes=32 time<1ms TTL=64

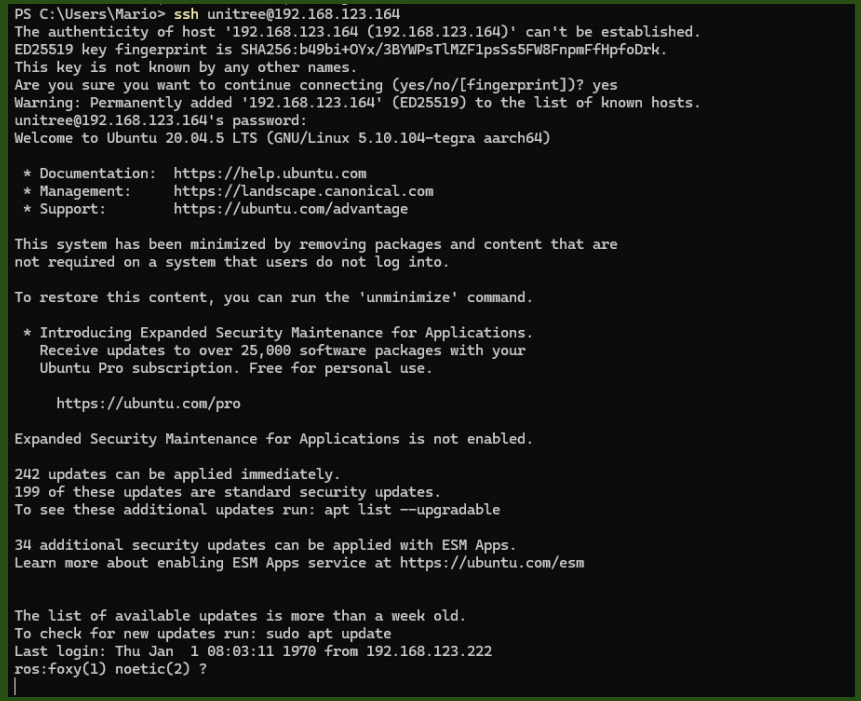
Ping statistics for 192.168.123.164:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
PS C:\Users\Mario> ping 192.168.123.161

Pinging 192.168.123.161 with 32 bytes of data:
Reply from 192.168.123.161: bytes=32 time<1ms TTL=64
Reply from 192.168.123.161: bytes=32 time<1ms TTL=64
Reply from 192.168.123.161: bytes=32 time<1ms TTL=64
Reply from 192.168.123.161: bytes=32 time=1ms TTL=64

Ping statistics for 192.168.123.161:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms
PS C:\Users\Mario>
```

Logging into the R1’s computer (backpack)

- This noted that the R1’s computer can apply 242 updates.



Logging into the R1's computer (backpack)

```
unitree@ubuntu:~$ whoami
unitree
unitree@ubuntu:~$ hostname
ubuntu
unitree@ubuntu:~$ echo $ROS_DISTRO

unitree@ubuntu:~$ echo $ROS_DISTRO

unitree@ubuntu:~$ date
Thu 22 Oct 2026 03:46:46 AM CST
unitree@ubuntu:~$ ip -br addr
lo               UNKNOWN        127.0.0.1/8 ::1/128
dummy0           DOWN
eth10            UP             192.168.123.164/24 fe80::4ebb:47ff:feab:e861/64
docker0          DOWN           172.17.0.1/16
eth0             DOWN
```

Logging into the R1's computer (backpack)

```
unitree@ubuntu:~$ ls -la ~
total 29320
drwxr-xr-x 31 unitree unitree     4096 Jan  1  1970 .
drwxr-xr-x  4 root    root        4096 Nov  7  2023 ..
-rw-rw-r--  1 unitree unitree        0 Dec 30  2025 a.out
-rw-------  1 unitree unitree    19356 Jan  1  1970 .bash_history
-rw-r--r--  1 unitree unitree      220 Sep  8  2022 .bash_logout
-rw-r--r--  1 unitree unitree     4287 Mar 11  2024 .bashrc
drwxrwxr-x  9 unitree unitree     4096 Apr 24 20:10 brainco_hand_service
-rw-rw-r--  1 unitree unitree 10900260 Jan  1  1970 brainco_hand_service.zip
drwxr-xr-x 16 unitree unitree     4096 Mar  4  2026 .cache
drwxr-xr-x  6 root    root        4096 Sep  8  2025 ch343ser_linux-main
-rw-rw-r--  1 unitree unitree   319523 Dec 30  2025 ch343ser_linux-main.zip
drwxrwxr-x  3 unitree unitree     4096 Sep  4  2023 .cmake
drwx------ 19 unitree unitree     4096 Jan  1  1970 .config
drwxrwxr-x  6 unitree unitree     4096 Nov  7  2023 cyclonedds_ws
drwx------  3 unitree unitree     4096 Nov  7  2023 .dbus
drwxr-xr-x  2 unitree unitree     4096 Nov  6  2023 Desktop
drwxr-xr-x  3 unitree unitree     4096 Mar 11  2024 Documents
drwxr-xr-x  3 unitree unitree     4096 Nov  7  2023 Downloads
-rw-rw-r--  1 unitree unitree  1566026 Feb  3  2026 'ebe920bb65ac486bb571b7fc429bd9ed (1).zip'
drwx------  3 unitree unitree     4096 Mar  4  2026 .gnupg
drwxrwxr-x  6 unitree unitree     4096 Feb  3  2026 h1_inspire_service
-rwxrwxr-x  1 unitree unitree     1338 Jan  1  1970 hand_set.sh
-rwxrwxrwx  1 root    root         145 Feb  3  2026 inspire_hand.sh
drwxrwxr-x  8 unitree unitree     4096 Mar  4  2026 linker_hand_service
drwxr-xr-x  4 unitree unitree     4096 Nov  3  2023 .local
drwx------  4 unitree unitree     4096 Sep  1  2023 .mozilla
drwxr-xr-x  2 unitree unitree     4096 Sep  8  2022 Music
-rwxr-xr-x  1 unitree unitree       87 Jun  5  2023 nomachine.sh
drwx------ 54 unitree unitree     4096 Jan  1  1970 .nx
drwxr-xr-x  2 unitree unitree     4096 Jan  1  1970 Pictures
-rw-r--r--  1 unitree unitree      807 Sep  8  2022 .profile
drwxr-xr-x  2 unitree unitree     4096 Sep  8  2022 Public
drwx------  2 unitree unitree     4096 Nov  7  2023 .qt
drwxrwxr-x  4 unitree unitree     4096 Jan  1  1970 .ros
drwxrwxr-x  2 unitree unitree     4096 Sep  4  2023 .rviz
drwx------  2 unitree unitree     4096 Nov  1  2023 .ssh
drwxrwxr-x  9 unitree unitree     4096 Feb  3  2026 stark-serialport-example
-rw-r--r--  1 unitree unitree        0 Sep  8  2022 .sudo_as_admin_successful
drwxr-xr-x  2 unitree unitree     4096 Sep  8  2022 Templates
drwxrwxr-x  3 unitree unitree     4096 Apr  3  2026 unitree
drwxrwxr-x 12 unitree unitree     4096 Apr  3  2026 unitree_sdk2
drwxrwxr-x 11 unitree unitree     4096 Feb  3  2026 unitree_sdk2-main
-rw-rw-r--  1 unitree unitree 17018603 Feb  3  2026 unitree_sdk2-main.zip
drwxr-xr-x  2 unitree unitree     4096 Sep  8  2022 Videos
-rw-------  1 unitree unitree    13202 Jan  1  1970 .viminfo
-rw-rw-r--  1 unitree unitree      245 Feb  3  2026 .wget-hsts
-rw-------  1 unitree unitree      312 Feb  3  2026 .Xauthority
-rw-r--r--  1 unitree unitree     4017 Sep  8  2022 .xsessionrc
unitree@ubuntu:~$
```

Signing into Unitree_SDK

```
unitree@ubuntu:~$ ls -la ~/unitree_sdk2
total 64
drwxrwxr-x 12 unitree unitree 4096 Apr  3  2026 .
drwxr-xr-x 31 unitree unitree 4096 Jan  1  1970 ..
drwxrwxr-x  6 unitree unitree 4096 Apr  3  2026 build
drwxrwxr-x  2 unitree unitree 4096 Apr  3  2026 cmake
-rw-rw-r--  1 unitree unitree 3012 Apr  3  2026 CMakeLists.txt
drwxrwxr-x  2 unitree unitree 4096 Apr  3  2026 .devcontainer
drwxrwxr-x 14 unitree unitree 4096 Apr  3  2026 example
drwxrwxr-x  8 unitree unitree 4096 Apr  3  2026 .git
drwxrwxr-x  3 unitree unitree 4096 Apr  3  2026 .github
-rw-rw-r--  1 unitree unitree   42 Apr  3  2026 .gitignore
drwxrwxr-x  3 unitree unitree 4096 Apr  3  2026 include
drwxrwxr-x  4 unitree unitree 4096 Apr  3  2026 lib
-rw-rw-r--  1 unitree unitree 1567 Apr  3  2026 LICENSE
drwxrwxr-x  5 unitree unitree 4096 Apr  3  2026 licenses
-rw-rw-r--  1 unitree unitree 1431 Apr  3  2026 README.md
drwxrwxr-x  4 unitree unitree 4096 Apr  3  2026 thirdparty
unitree@ubuntu:~$ find ~/unitree_sdk2 ~/unitree_sdk2-main -maxdepth 5 -type f \( -iname '*r1*' -o -ina
me '*arm*' \) | head -n 50
/home/unitree/unitree_sdk2/build/bin/g1_arm_action_example
/home/unitree/unitree_sdk2/build/bin/r1_loco_client
/home/unitree/unitree_sdk2/build/bin/g1_arm7_sdk_dds_example
/home/unitree/unitree_sdk2/build/bin/g1_arm5_sdk_dds_example
/home/unitree/unitree_sdk2/build/bin/h1_2_arm_sdk_dds_example
/home/unitree/unitree_sdk2/build/bin/r1_ankle_swing_example
/home/unitree/unitree_sdk2/build/bin/h1_arm_sdk_dds_example
/home/unitree/unitree_sdk2/build/bin/g1_dual_arm_example
/home/unitree/unitree_sdk2/example/h1/high_level/h1_arm_sdk_dds_example.cpp
/home/unitree/unitree_sdk2/example/h1/high_level/h1_2_arm_sdk_dds_example.cpp
/home/unitree/unitree_sdk2/example/r1/high_level/r1_loco_client_example.cpp
/home/unitree/unitree_sdk2/example/r1/low_level/r1_ankle_swing_example.cpp
/home/unitree/unitree_sdk2/example/g1/high_level/g1_arm7_sdk_dds_example.cpp
/home/unitree/unitree_sdk2/example/g1/high_level/g1_arm5_sdk_dds_example.cpp
/home/unitree/unitree_sdk2/example/g1/high_level/g1_arm_action_example.cpp
/home/unitree/unitree_sdk2/example/g1/low_level/g1_dual_arm_example.cpp
/home/unitree/unitree_sdk2-main/example/h1/high_level/h1_arm_sdk_dds_example.cpp
/home/unitree/unitree_sdk2-main/example/h1/high_level/h1_2_arm_sdk_dds_example.cpp
/home/unitree/unitree_sdk2-main/example/g1/high_level/g1_arm7_sdk_dds_example.cpp
/home/unitree/unitree_sdk2-main/example/g1/high_level/g1_arm5_sdk_dds_example.cpp
/home/unitree/unitree_sdk2-main/example/g1/high_level/g1_arm_action_example.cpp
/home/unitree/unitree_sdk2-main/example/g1/low_level/g1_dual_arm_example.cpp
```

Finding R1-Specific SDK commands

We can see that the SDK's last update was in March 2026, and we know the upper body R1 SDK scripts were added to the SDK environment on GitHub on August 2026.

Therefore we must add our own SDK upper body R1 scripts.

Also, since these scripts developed by Unitree most likely use all joints, we must make our own customized script that keeps lower body joints not moving (as the R1 is currently on the bracket; also just having upper body control is useful for future hand joint development).

```
unitree@ubuntu:~$ ps -eo pid,comm,args | grep -Ei 'unitree|sport|arm|dds' | grep -v grep | head -n 30
   1096 master_service  /unitree/module/master_service/master_service
   1098 ota_pipe_servic /unitree/ota/pipe/ota_pipe_service
   2170 sshd            sshd: unitree [priv]
   2357 sshd            sshd: unitree@pts/0
unitree@ubuntu:~$ git -C ~/unitree_sdk2 log -1 --format='%h %cs %s'
fa925bf 2026-03-25 Merge branch 'dev/add_r1_ankle_swing_example'
unitree@ubuntu:~$ git -C ~/unitree_sdk2 status --short
 T thirdparty/lib/aarch64/libddsc.so.0
 T thirdparty/lib/aarch64/libddscxx.so.0
 T thirdparty/lib/x86_64/libddsc.so.0
 T thirdparty/lib/x86_64/libddscxx.so.0
unitree@ubuntu:~$ ls -la ~/unitree_sdk2/example/r1/high_level/
total 16
drwxrwxr-x 2 unitree unitree 4096 Apr  3  2026 .
drwxrwxr-x 4 unitree unitree 4096 Apr  3  2026 ..
-rw-rw-r-- 1 unitree unitree 4500 Apr  3  2026 r1_loco_client_example.cpp
unitree@ubuntu:~$ cat ~/unitree_sdk2/example/r1/CMakeLists.txt
cmake_minimum_required(VERSION 3.10)

add_executable(r1_loco_client high_level/r1_loco_client_example.cpp)
target_link_libraries(r1_loco_client unitree_sdk2)

add_executable(r1_ankle_swing_example low_level/r1_ankle_swing_example.cpp)
target_link_libraries(r1_ankle_swing_example unitree_sdk2)
unitree@ubuntu:~$
```

Download Github SDK arm repo's without wifi

```
PS C:\Users\Mario> cd "$env:USERPROFILE\Downloads"
PS C:\Users\Mario\Downloads> curl.exe -L --fail "https://github.com/unitreerobotics/unitree_sdk2/archive/refs/heads/main
.tar.gz" -o unitree_sdk2-main.tar.gz
  % Total    % Received % Xferd  Average Speed  Time    Time    Time   Current
                                 Dload  Upload  Total   Spent   Left   Speed
  0      0   0      0   0      0      0      0           0              0
100 16.29M   0 16.29M   0      0  5.14M      0          00:03          5.49M
PS C:\Users\Mario\Downloads> scp .\unitree_sdk2-main.tar.gz unitree@192.168.123.164:/home/unitree/r1_custom/
unitree@192.168.123.164's password:
unitree_sdk2-main.tar.gz                                                    100%   16MB 114.0MB/s   00:00
PS C:\Users\Mario\Downloads>
```

```
unitree@ubuntu:~$ ls -la ~/r1_custom
total 16708
drwxrwxr-x  5 unitree unitree     4096 Oct 22 04:15 .
drwxr-xr-x 32 unitree unitree     4096 Oct 22 04:09 ..
drwxrwxr-x  2 unitree unitree     4096 Oct 22 04:09 build
drwxrwxr-x  2 unitree unitree     4096 Oct 22 04:09 config
drwxrwxr-x  6 unitree unitree     4096 Oct 22 04:09 src
-rw-rw-r--  1 unitree unitree 17085689 Oct 22 04:15 unitree_sdk2-main.tar.gz
unitree@ubuntu:~$ tar -xzf ~/r1_custom/unitree_sdk2-main.tar.gz -C ~/r1_custom
unitree@ubuntu:~$ mv ~/r1_custom/unitree_sdk2-main ~/r1_custom/unitree_sdk2
unitree@ubuntu:~$ ls -la ~/r1_custom/unitree_sdk2/example/r1/high_level/
total 36
drwxrwxr-x 2 unitree unitree 4096 Oct 15 18:02 .
drwxrwxr-x 5 unitree unitree 4096 Oct 15 18:02 ..
-rw-rw-r-- 1 unitree unitree 9575 Oct 15 18:02 r1_arm_action_example.cpp
-rw-rw-r-- 1 unitree unitree 6104 Oct 15 18:02 r1_arm_sdk_dds_example.cpp
-rw-rw-r-- 1 unitree unitree 4500 Oct 15 18:02 r1_loco_client_example.cpp
unitree@ubuntu:~$ ls -la ~/r1_custom
total 16712
drwxrwxr-x  6 unitree unitree     4096 Oct 22 04:16 .
drwxr-xr-x 32 unitree unitree     4096 Oct 22 04:09 ..
drwxrwxr-x  2 unitree unitree     4096 Oct 22 04:09 build
drwxrwxr-x  2 unitree unitree     4096 Oct 22 04:09 config
drwxrwxr-x  6 unitree unitree     4096 Oct 22 04:09 src
drwxrwxr-x 10 unitree unitree     4096 Oct 15 18:02 unitree_sdk2
-rw-rw-r--  1 unitree unitree 17085689 Oct 22 04:15 unitree_sdk2-main.tar.gz
unitree@ubuntu:~$
```

Compiling R1 Arm SDK scripts

```
unitree@ubuntu:~$ cmake --version
cmake version 3.16.3

CMake suite maintained and supported by Kitware (kitware.com/cmake).
unitree@ubuntu:~$ g++ --version
g++ (Ubuntu 9.4.0-1ubuntu1~20.04.2) 9.4.0
Copyright (C) 2019 Free Software Foundation, Inc.
This is free software; see the source for copying conditions.  There is NO
warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.

unitree@ubuntu:~$ cmake -S ~/r1_custom/unitree_sdk2 \
>       -B ~/r1_custom/unitree_sdk2/build
-- The C compiler identification is GNU 9.4.0
-- The CXX compiler identification is GNU 9.4.0
-- Check for working C compiler: /usr/bin/cc
-- Check for working C compiler: /usr/bin/cc -- works
-- Detecting C compiler ABI info
-- Detecting C compiler ABI info - done
-- Detecting C compile features
-- Detecting C compile features - done
-- Check for working CXX compiler: /usr/bin/c++
-- Check for working CXX compiler: /usr/bin/c++ -- works
-- Detecting CXX compiler ABI info
-- Detecting CXX compiler ABI info - done
-- Detecting CXX compile features
-- Detecting CXX compile features - done
-- Setting build type to 'Release' as none was specified.
-- Current system architecture: aarch64
-- Looking for pthread.h
-- Looking for pthread.h - found
-- Performing Test CMAKE_HAVE_LIBC_PTHREAD
-- Performing Test CMAKE_HAVE_LIBC_PTHREAD - Failed
-- Looking for pthread_create in pthreads
-- Looking for pthread_create in pthreads - not found
-- Looking for pthread_create in pthread
-- Looking for pthread_create in pthread - found
-- Found Threads: TRUE
-- Importing: /home/unitree/r1_custom/unitree_sdk2/thirdparty/lib/aarch64/libddsc.so
-- Importing: /home/unitree/r1_custom/unitree_sdk2/thirdparty/lib/aarch64/libddscxx.so
-- Unitree SDK library found at: /home/unitree/r1_custom/unitree_sdk2/lib/aarch64/libunitree_sdk2.a
-- Importing: /home/unitree/r1_custom/unitree_sdk2/lib/aarch64/libunitree_sdk2.a
-- Found Boost: /usr/lib/aarch64-linux-gnu/cmake/Boost-1.71.0/BoostConfig.cmake (found version "1.71.0
") found components: program_options
-- Found yaml-cpp version 0.6.2
-- Configuring done
-- Generating done
-- Build files have been written to: /home/unitree/r1_custom/unitree_sdk2/build
```

```
unitree@ubuntu:~$ cmake --build ~/r1_custom/unitree_sdk2/build \
>       --target r1_arm_action_example r1_arm_sdk_dds_example \
>       -j2
Scanning dependencies of target r1_arm_action_example
[ 50%] Building CXX object example/r1/CMakeFiles/r1_arm_action_example.dir/high_level/r1_arm_action_example.cpp.o
[100%] Linking CXX executable ../../bin/r1_arm_action_example
[100%] Built target r1_arm_action_example
Scanning dependencies of target r1_arm_sdk_dds_example
[100%] Building CXX object example/r1/CMakeFiles/r1_arm_sdk_dds_example.dir/high_level/r1_arm_sdk_dds_example.cpp.o
[100%] Linking CXX executable ../../bin/r1_arm_sdk_dds_example
[100%] Built target r1_arm_sdk_dds_example
unitree@ubuntu:~$ ls -lh ~/r1_custom/unitree_sdk2/build/bin/r1_arm*
-rwxrwxr-x 1 unitree unitree 3.5M Oct 22 04:23 /home/unitree/r1_custom/unitree_sdk2/build/bin/r1_arm_action_example
-rwxrwxr-x 1 unitree unitree 2.9M Oct 22 04:23 /home/unitree/r1_custom/unitree_sdk2/build/bin/r1_arm_sdk_dds_example
```

- Important Notes:
- Cmake Version: 3.16.3
- GCC/G++ Version: 9.4.0
- ARM 64 Architecture detected
- Cyclone DDS libraries, Unitree SDK2 library found
- R1 Arm Action Example and R1 Arm DDS Example Compiles, therefore the build was successful.

Query the R1's Available Arm Actions

```
unitree@ubuntu:~$ ~/r1_custom/unitree_sdk2/build/bin/r1_arm_action_example eth10 --list
 --- Unitree Robotics ---
     R1 Arm Action Example

Available actions:
[[{"id": 99, "name": "release_arm"}, {"id": 11, "name": "blow_kiss_with_both_hands"}, {"id": 12, "name
": "blow_kiss_with_left_hand"}, {"id": 13, "name": "blow_kiss_with_right_hand"}, {"id": 15, "name": "b
oth_hands_up"}, {"id": 17, "name": "clamp"}, {"id": 18, "name": "high_five"}, {"id": 19, "name": "hug"
}, {"id": 22, "name": "refuse"}, {"id": 23, "name": "right_hand_up"}, {"id": 24, "name": "ultraman_ray
"}, {"id": 25, "name": "wave_under_head"}, {"id": 26, "name": "wave_above_head"}, {"id": 27, "name": "
shake_hand"}, {"id": 28, "name": "box_left_hand_win"}, {"id": 29, "name": "box_right_hand_win"}, {"id"
: 30, "name": "box_both_hand_win"}, {"id": 31, "name": "extend_right_arm_forward"}, {"id": 33, "name":
 "right_hand_on_heart"}, {"id": 34, "name": "both_hands_up_deviate_right"}, {"id": 35, "name": "emphas
ize"}, {"id": 36, "name": "forward_push"}], []]
unitree@ubuntu:~$
```

We can see there are several commands that are already available.
Though for now we will not use these commands due to the fact that

Making a Custom Command for Joint Monitoring

```
unitree@ubuntu:~$ cat ~/r1_custom/src/upper_body/r1_joint_monitor.cpp
#include <array>
#include <chrono>
#include <iostream>
#include <memory>
#include <mutex>
#include <thread>

#include "unitree/robot/channel/channel_factory.hpp"
#include "unitree/dds_wrapper/robots/g1/g1.h"
#include "unitree/dds_wrapper/robots/r1/defines.h"

using unitree::robot::r1::JointIndex;

struct Joint {
    const char* name;
    JointIndex index;
};

int main() {
    unitree::robot::ChannelFactory::Instance()->Init(0, "eth10");

    auto state =
        std::make_shared<unitree::robot::g1::subscription::LowState>();

    const std::array<Joint, 13> joints = {{
        {"Left Shoulder Pitch", JointIndex::LeftShoulderPitch},
        {"Left Shoulder Roll",  JointIndex::LeftShoulderRoll},
        {"Left Shoulder Yaw",   JointIndex::LeftShoulderYaw},
        {"Left Elbow",          JointIndex::LeftElbow},
        {"Left Wrist Roll",     JointIndex::LeftWristRoll},
        {"Right Shoulder Pitch",JointIndex::RightShoulderPitch},
        {"Right Shoulder Roll", JointIndex::RightShoulderRoll},
        {"Right Shoulder Yaw",  JointIndex::RightShoulderYaw},
        {"Right Elbow",         JointIndex::RightElbow},
        {"Right Wrist Roll",    JointIndex::RightWristRoll},
        {"Waist Yaw",           JointIndex::WaistYaw},
        {"Head Pitch",          JointIndex::HeadPitch},
        {"Head Yaw",            JointIndex::HeadYaw}
    }};

    std::cout << "Waiting for R1 joint feedback...\n";

    for (int i = 0; i < 50 && state->isTimeout(); ++i)
        std::this_thread::sleep_for(std::chrono::milliseconds(100));

    if (state->isTimeout()) {
        std::cerr << "No joint feedback within 5 seconds.\n";
        return 1;
    }

    for (int sample = 0; sample < 3; ++sample) {
        {
            std::lock_guard<std::mutex> lock(state->mutex_);

            if (state->isTimeout()) {
                std::cerr << "Joint feedback timed out.\n";
                return 2;
            }

            const auto& motors = state->msg_.motor_state();

            std::cout << "\n--- Sample " << sample + 1 << " ---\n";

            for (const auto& joint : joints) {
                auto index = static_cast<size_t>(joint.index);

                if (index < motors.size())
                    std::cout << joint.name << ": "
                              << motors.at(index).q() << " rad\n";
            }
        }

        std::this_thread::sleep_for(std::chrono::seconds(1));
    }

    return 0;
}
```

Joint Feedback from Physical Robot



```
unitree@ubuntu:~$ ~/r1_custom/build/r1_joint_monitor
Waiting for R1 joint feedback...

--- Sample 1 ---
Left Shoulder Pitch: 0.0602447 rad
Left Shoulder Roll: 0.00254665 rad
Left Shoulder Yaw: 0.553929 rad
Left Elbow: 1.32963 rad
Left Wrist Roll: 0.723065 rad
Right Shoulder Pitch: 0.0717735 rad
Right Shoulder Roll: -0.0044821 rad
Right Shoulder Yaw: -0.00887085 rad
Right Elbow: 1.14825 rad
Right Wrist Roll: 0.00222528 rad
Waist Yaw: -0.114755 rad
Head Pitch: 0.0152439 rad
Head Yaw: -0.136897 rad

--- Sample 2 ---
Left Shoulder Pitch: 0.0602507 rad
Left Shoulder Roll: 0.00254066 rad
Left Shoulder Yaw: 0.553944 rad
Left Elbow: 1.32961 rad
Left Wrist Roll: 0.72308 rad
Right Shoulder Pitch: 0.0717675 rad
Right Shoulder Roll: -0.00447611 rad
Right Shoulder Yaw: -0.00888599 rad
Right Elbow: 1.14826 rad
Right Wrist Roll: 0.00224042 rad
Waist Yaw: -0.114749 rad
Head Pitch: 0.0152439 rad
Head Yaw: -0.136902 rad

--- Sample 3 ---
Left Shoulder Pitch: 0.0602447 rad
Left Shoulder Roll: 0.00254665 rad
Left Shoulder Yaw: 0.553913 rad
Left Elbow: 1.32961 rad
Left Wrist Roll: 0.72305 rad
Right Shoulder Pitch: 0.0717735 rad
Right Shoulder Roll: -0.0044821 rad
Right Shoulder Yaw: -0.00887085 rad
Right Elbow: 1.14825 rad
Right Wrist Roll: 0.00224042 rad
Waist Yaw: -0.114755 rad
Head Pitch: 0.0152439 rad
Head Yaw: -0.136896 rad
```



```
unitree@ubuntu:~$ ~/r1_custom/build/r1_joint_monitor
Waiting for R1 joint feedback...

--- Sample 1 ---
Left Shoulder Pitch: 0.0602267 rad
Left Shoulder Roll: 0.00254665 rad
Left Shoulder Yaw: 0.553898 rad
Left Elbow: 1.32961 rad
Left Wrist Roll: 0.72302 rad
Right Shoulder Pitch: 0.0717675 rad
Right Shoulder Roll: -0.00443416 rad
Right Shoulder Yaw: -0.00885571 rad
Right Elbow: 1.14825 rad
Right Wrist Roll: 0.00224042 rad
Waist Yaw: -0.114743 rad
Head Pitch: -0.00906764 rad
Head Yaw: 2.00713 rad

--- Sample 2 ---
Left Shoulder Pitch: 0.0602327 rad
Left Shoulder Roll: 0.00252867 rad
Left Shoulder Yaw: 0.553913 rad
Left Elbow: 1.32963 rad
Left Wrist Roll: 0.723035 rad
Right Shoulder Pitch: 0.0717795 rad
Right Shoulder Roll: -0.00444016 rad
Right Shoulder Yaw: -0.00887085 rad
Right Elbow: 1.14825 rad
Right Wrist Roll: 0.00225556 rad
Waist Yaw: -0.114737 rad
Head Pitch: -0.00908278 rad
Head Yaw: 2.00712 rad

--- Sample 3 ---
Left Shoulder Pitch: 0.0602267 rad
Left Shoulder Roll: 0.00254665 rad
Left Shoulder Yaw: 0.553944 rad
Left Elbow: 1.32961 rad
Left Wrist Roll: 0.72305 rad
Right Shoulder Pitch: 0.0717795 rad
Right Shoulder Roll: -0.00443416 rad
Right Shoulder Yaw: -0.00888599 rad
Right Elbow: 1.14823 rad
Right Wrist Roll: 0.00225556 rad
Waist Yaw: -0.114743 rad
Head Pitch: -0.00908278 rad
Head Yaw: 2.00713 rad
```

We can see when moving the head of the robot, the Head Pitch and Head Yaw angles change, showing that the Joint Monitoring Script works.

Turning off ssh

```
unitree@ubuntu:~$ sudo shutdown -h now
[sudo] password for unitree:
Connection to 192.168.123.164 closed by remote host.
Connection to 192.168.123.164 closed.
PS C:\Users\Mario>
```

Next Steps: Customized Arm SDK command

- We want to customize the SDK file to be able to move the upper body components while ensuring lower body components are ignored.
- So we will send our own target joint positions through DDS

References

- https://www.unitree.com/app/R1
- https://docs.westonrobot.com/robot/humanoid/r1
- https://github.com/unitreerobotics/xr_teleoperate/wiki/Network
- https://github.com/unitreerobotics/unitree_sdk2
- https://github.com/unitreerobotics/unitree_sdk2/blob/main/example/r1/high_level/r1_arm_action_example.cpp
- https://github.com/unitreerobotics/unitree_sdk2/blob/main/example/r1/high_level/r1_arm_sdk_dds_example.cpp
- https://support.unitree.com/home/en/R1_developer/edu_backpack